

SOLAR HOME PROJECT

Science Exhibition Model Report

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Project Topic: Solar Home System

Introduction

Solar home systems basically consist of solar panels which are built on the roof according to requirements and can be used to fulfill the energy needed to meet a household's energy demand, fulfilling basic electric needs.

Such homes are especially useful for remote locations where electricity by the grid is not possible. Solar panels can provide power for both business and residential use. In both cases, photovoltaic panels are mounted on the roofs to capture as much sunlight as possible and create as much power.

Problem Statement

People living in rural areas and off-grid regions where electricity connections are not possible find it difficult to perform different activities at night.

Proposed Solution

My proposed model is about a solar home. A solar home system uses solar panels, batteries, and other components to provide electricity to homes not connected to the electricity grid, replacing traditional resources like kerosene lamps for lighting.

Materials Needed

• Wires & Cellotape

• Real Solar Panel

• LED String Lights & Bulbs

• Hot Glue Gun / Adhesive

• Cardboard Pieces & Charts

• Scissors, Straws & Box Decoration

• Artificial Trees & Toys (for realistic model look)

Construction and Working

Take a plywood piece and cover it with a chart to make the base of the model. Cut cardboard pieces to design the structure of the house. Place the solar panel at the top on the roof of the house. Install electric wiring connecting the solar panel to the battery system and inverter, using small LED lights to demonstrate the working electrical energy generated from the solar panel.

Benefits

- **Pollution Reduction:** Solar homes are useful in reducing indoor and outdoor air pollution, thereby improving health and minimizing environmental degradation.
- **Grid-Free Access:** They provide the capability and convenience of working and studying at night in areas completely free from conventional electrical grids.

Conclusion

The use of solar homes avoids greenhouse gas emissions by drastically reducing reliance on conventional energy resources such as kerosene, dry cell batteries, or diesel generators traditionally used for electricity generation.